Diffusion & Mixing

Why microscale mixing is slow — a Semitree cheat sheet

With no turbulence at low Reynolds number, laminar streams mix only by diffusion. Time scales with the square of distance, so shrinking the mixing distance is what speeds mixing up.

Diffusion time (estimate) — t

$t \approx L^2 / (2 \cdot D)$

order-of-magnitude characteristic time, not an exact profile

Diffusion length — L

$L \approx \sqrt{2 \cdot D \cdot t}$

how far a species spreads in time t

Péclet number — Pe

$Pe = v \cdot L / D$

advection vs. diffusion

Stokes–Einstein — D

$D = k_B \cdot T / (6 \cdot \pi \cdot \mu \cdot r)$

estimate D for a spherical particle

Mixing length — Lmix

$L_{mix} \approx Pe \cdot w$

channel length needed to mix across width w (order)

Variables

t time (s)	**L** distance (m)	**D** diffusion coeff. (m^2/s)
v mean velocity (m/s)	**kB** Boltzmann const. (J/K)	**T** temperature (K)
μ dynamic viscosity (Pa·s)	**r** particle radius (m)	**w** channel width (m)

Standard, published equations. Always verify against primary sources for critical work.